

(a) Mg^{2+} (b) Ti^{3+} (c) V^{3+} (d) Fe^{2+}

6. The statement that is not correct for the periodic classification of element is [1992 - 1 Mark]

- (a) The properties of elements are the periodic functions of their atomic numbers
- (b) Non-metallic elements are lesser in number than metallic elements
- (c) The first ionisation energies of elements along a period do not vary in a regular manner with increase in atomic number
- (d) For transition elements the d -subshells are filled with electrons monotonically with increase in atomic number.



3

Numeric / New Stem Based Questions

7. If IUPAC name of an element is "Unununnium" then the element belongs to n^{th} group of Periodic table. The value of n is _____. [Main Jan. 30, 2024 (I)]



6

MCQs with One or More than One Correct Answer

8. The statements that **are true** for the long form of the periodic table are : [1988 - 1 Mark]

- (a) it reflects the sequence of filling the electrons in the order of sub-energy level s , p , d and f .
- (b) it helps to predict the stable valency states of the elements
- (c) it reflects trends in physical and chemical properties of the elements
- (d) it helps to predict the relative ionicity of the bond between any two elements.

2. The electron affinity value are negative for :
[Main April 6, 2024 (I)]

A. $\text{Be} \rightarrow \text{Be}^-$; B. $\text{N} \rightarrow \text{N}^-$; C. $\text{O} \rightarrow \text{O}_2^-$; D. $\text{Na} \rightarrow \text{Na}^-$
E. $\text{Al} \rightarrow \text{Al}^-$

Choose the most appropriate answer from the options given below :

- (a) D and E only (b) A, B, D and E only
(c) A and D only (d) A, B and C only

3. Evaluate the following statements related to group 14 elements for their correctness. [Main April 6, 2024 (II)]

- (A) Covalent radius decreases down the group from C to Pb in a regular manner.
(B) Electronegativity decreases from C to Pb down the group gradually.
(C) Maximum covalence of C is 4 whereas other elements can expand their covalence due to presence of d orbitals.
(D) Heavier elements do not form $p\pi-p\pi$ bonds.
(E) Carbon can exhibit negative oxidation states.

Choose the **correct** answer from the options given below:

- (a) (C), (D) and (E) Only
(b) (A) and (B) Only
(c) (A), (B) and (C) Only
(d) (C) and (D) Only

43.

Among the statements (I - IV), the **correct** ones are :

[Main Sep. 03, 2020 (II)]

- (I) Be has smaller atomic radius compared to Mg.
- (II) Be has higher ionization enthalpy than Al.
- (III) Charge/radius ratio of Be is greater than that of Al.
- (IV) Both Be and Al form mainly covalent compounds.

- (a) (I), (II) and (IV)
- (b) (I), (III) and (IV)
- (c) (II), (III) and (IV)
- (d) (I), (II) and (III)

54. Within each pair of elements F & Cl, S & Se, and Li & Na, respectively, the elements that release more energy upon an electron gain are: **[Main Jan. 07, 2020 (II)]**

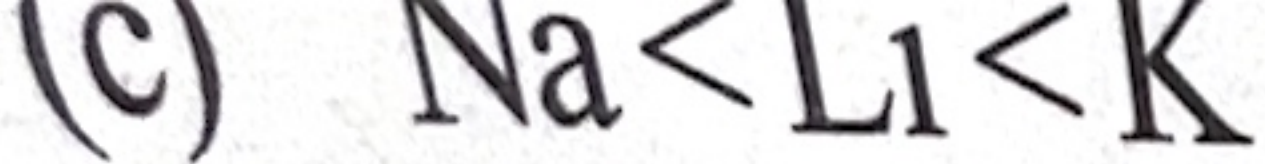
(a) Cl, Se and Na

(b) Cl, S and Li

~~(c) F, S and Li~~

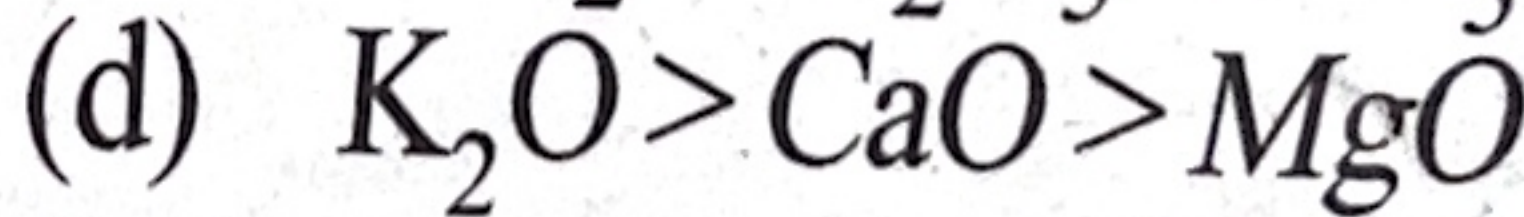
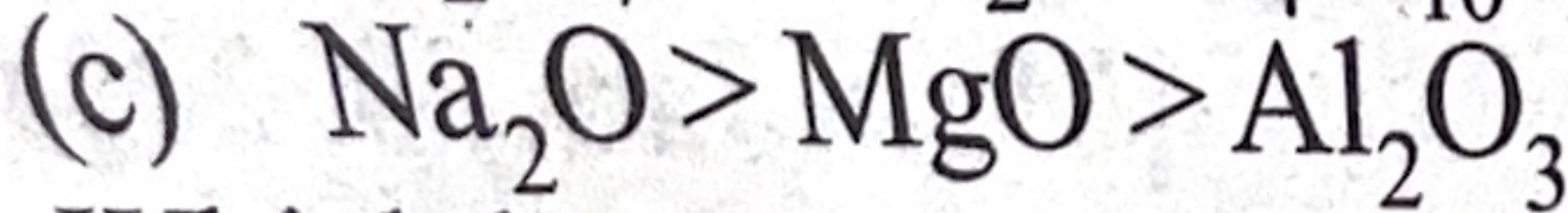
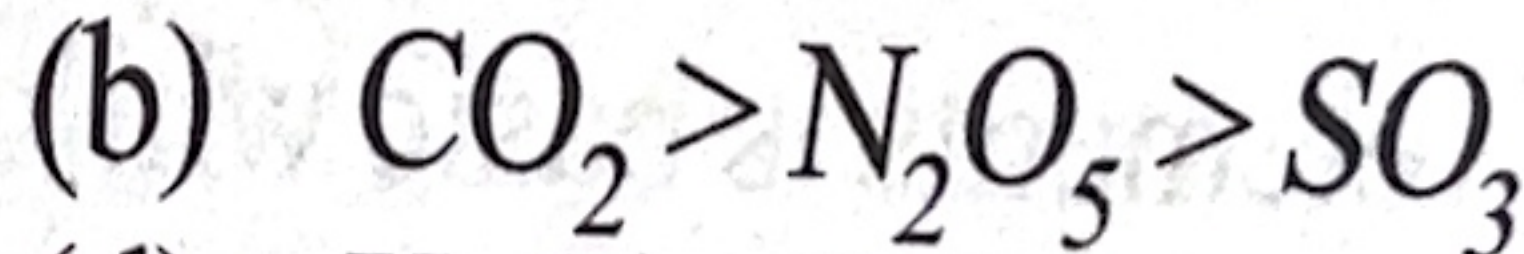
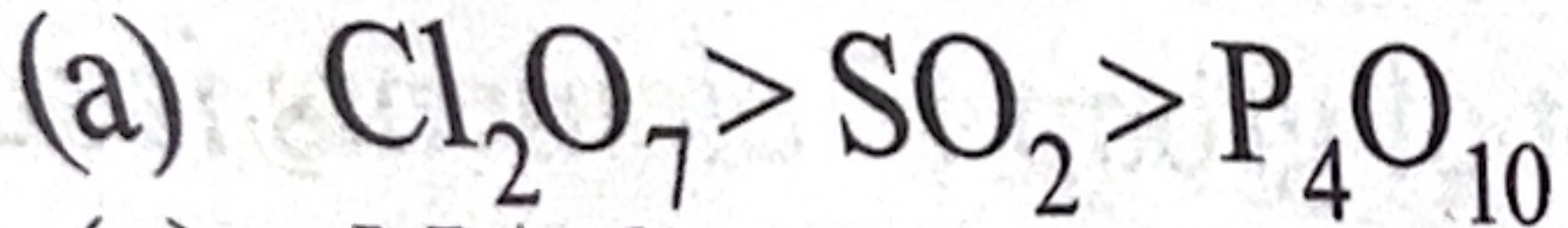
(d) F, Se and Na

75



The correct order of acidic strength is

[2000S]



76

Which has most stable ...

[Main Feb. 1, 2024 (II)]

106. **Statements – I:** Along the period, the chemical reactivity of the elements gradually increases from group 1 to group 18.

Statement – II: The nature of oxides formed by group 1 elements is basic while that of group 17 elements is acidic.

[Main Jan. 30, 2024 (II)]

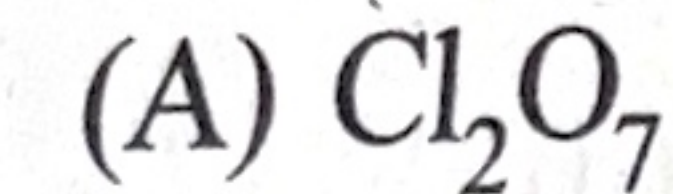
107. **Statement I :** The electronegativity of group 14 elements from Si to Pb, gradually decreases.

Statement II : Group 14 contains non-metallic, metallic, as well as metalloid elements. [Main Jan. 29, 2024 (I)]

102. Match List-I with List-II. [Main June 28, 2022 (II)]

List-I (Oxide)

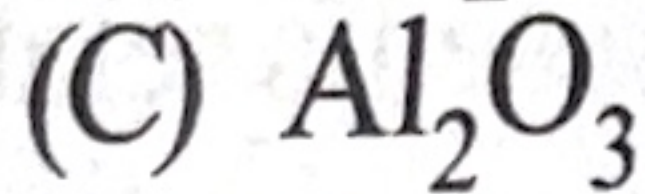
List-II (Nature)



(I) Amphoteric



(II) Basic



(III) Neutral



(IV) Acidic

Choose the correct answer from the options given below:

(a) (A) – (IV), (B) – (III), (C) – (I), (D) – (II)

(b) (A) – (IV), (B) – (II), (C) – (I), (D) – (III)

(c) (A) – (II), (B) – (IV), (C) – (III), (D) – (I)

(d) (A) – (I), (B) – (II), (C) – (III), (D) – (IV)